# Ayush Singh

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## Summary

Software Engineer III (L4) at Google (Munich) with 4+ years of experience architecting planet-scale distributed systems, low-latency C++/Java RPC frameworks, fault-tolerant distributed file systems (100K+ services), and real-time streaming pipelines. Proven track record of high-velocity technical leadership: defaulted microservice architecture across ~5,200 production nodes, delivered $2.5M+ quarterly cost savings, awarded a VP of Engineering Spot Bonus, earned fast-track promotion to L4 in 15 months, and certified with 3× Google Readability (C++, TypeScript, API Design; 1,040+ code reviews).

## Professional Experience

**Software Engineer III** November 2024 – Present
*Google, Munich, Germany* *Distributed Systems, Core C++/Java Infrastructure, System Design*

- Engineered high-throughput, low-latency request-path processors within **Google's core Java backend framework** (powering all Google products) and **core C++ Search serving engine**, operating on the critical serving path for billions of daily users.
- Authored and standardized the organization-wide microservice architecture defaulted across ~**5,200 production server nodes** (210+ modules across Search, YouTube, and Maps), reducing signal-processing complexity by **40%** and retiring **1,300+ lines of legacy code**.
- Built a parallel stream shadow-diffing framework across **YouTube, Google Play, Search, and Payments** achieving **100% live cut-over parity** across billions of RPCs, and migrated **7 mission-critical identity pipelines** to distributed batch compute infrastructure.
- Spearheaded end-to-end **UK Online Safety Act** and **US AppStore Age Verification** distributed backend rollouts across Web, Android, and iOS under an aggressive 4-week timeline across 5+ engineering and legal teams, mitigating multi-million dollar regulatory exposure and earning a **VP of Engineering Spot Bonus**.

**Software Engineer II → III** July 2022 – November 2024
*Google, India* *Distributed Storage Systems, Observability, Backend Infrastructure*

- Engineered fault-tolerant storage and replication for **Google's internal HDFS-like distributed file system**, optimizing deployment and distribution of mission-critical binary assets across **100,000+ internal production services**.
- Hardened distributed storage reliability by shipping server-side admission control and load-shedding, bounded semaphore RPC concurrency limiting in the **FUSE client daemon**, and migrating distributed snapshot locks off centralized **Paxos lock services** to eliminate lock contention.
- Architected a highly available observability and **automated SLA/SLO compliance enforcement platform** correlating distributed traces and time-series metrics across third-party and internal systems, delivering **$2.5M+ in quarterly cost savings**.
- Co-architected a real-time **OpenTelemetry** distributed trace collector over **gRPC-Web** and an end-to-end encrypted **Go static asset gateway** for automated cross-cluster storage migration.
- Led development of Google-wide Chrome third-party cookie deprecation impact analysis infrastructure securing **CEO/executive approval**, co-authored 3 org-wide Technical Design Documents (TDDs), and earned **fast-track promotion to SWE III (L4) within 15 months**.

**Software Engineer Intern** January 2022 – June 2022
*Komprise, India* *Automation, Python, CI/CD*

- Developed automated testing framework enabling parallel test execution across distributed infrastructure, improving bug detection coverage by 60%.
- Optimized Build Verification Test (BVT) suite through multi-threading implementation, achieving **87.5% reduction** in execution time and accelerating CI/CD pipeline throughput.

**Data Engineer Intern** May 2021 – July 2021
*Accenture, India* *Data Engineering, Cloud, Business Intelligence*

- Architected end-to-end Business Intelligence dashboard for E-Commerce analytics with real-time data visualization serving 50+ stakeholders.
- Designed optimized ETL data pipeline for high-volume data transfer (10TB+) from AWS S3 staging to Snowflake Data Warehouse.
- Developed NLP-powered conversational AI chatbot using Rasa for automated data insights extraction, reducing manual reporting time by 70%.

**Software Engineer Intern** October 2020 – January 2021
*Tookitaki, Singapore* *Real-time Data Processing, Distributed Systems*

- Engineered real-time streaming data pipeline using Apache Spark Structured-Streaming with ActiveMQ integration for high-throughput message processing.

- Achieved **94% latency reduction** (850ms to <50ms) through custom JMS-Client implementation with optimized connection pooling.
- Implemented fault-tolerant message recovery mechanism ensuring exactly-once delivery semantics in distributed streaming application.

## Education

**National Institute of Technology, Agartala** August 2018 – May 2022
*Bachelor of Technology in Electrical Engineering* *CGPA: 9.32/10*

## Technical Skills

**Programming Languages**: C++, Java, Go, TypeScript, Python, Scala, Kotlin, JavaScript, SQL
**Distributed Systems & Backend**: gRPC, Protobuf, Distributed File Systems (HDFS, FUSE), Paxos/Raft Consensus, Apache Spark, Apache Kafka, ActiveMQ, OpenTelemetry, Node.js, FastAPI
**Databases & Storage**: Spanner, Snowflake, PostgreSQL, MongoDB, MySQL, Redis, HDFS
**Cloud & DevOps**: Google Cloud Platform (GCP), AWS (S3, EC2), Docker, Kubernetes, Git, CI/CD, Jenkins, Terraform
**Machine Learning & AI Systems**: Gemini API, WebLLM, WebGPU, Multi-Agent Systems, TensorFlow, PyTorch, NLP

## Achievements & Recognition

• Certified with **3× Google Readability** (C++, TypeScript, and API Design) with **1,040+ production code reviews**, a **VP of Engineering Spot Bonus**, and **7 peer bonuses** across Google engineering teams.
• Institute **Rank 1** on GeeksForGeeks (among 1,180+ students) with worldwide **Rank 72** (among 100,000+) in Data Structures & Algorithms.
• Selected as **1 of 50 students** across India for **Google ExploreML 2019** machine learning program.
• Selected as **1 of 11 students** for the prestigious **Indian Academy of Sciences FASTSF Fellowship 2021**.
• Established first Google Developer Student Club at NIT Agartala as Google-appointed Lead (1 of 179 leads selected across India).

## Technical Projects

**YTM AI Host – AI-Powered Music Radio Jockey** GitHub
*TypeScript, Gemini API, WebLLM, WebGPU, Python, Chrome Extensions*

- Developed AI-powered Radio Jockey Chrome extension for YouTube Music, delivering dynamic real-time introductions using large language models.
- Integrated Google Gemini API and WebLLM for on-device LLM inference with multiple Text-to-Speech engines (Chrome TTS, Kokoro JS, XTTS v2).
- Implemented intelligent caching and pre-fetching algorithms achieving sub-100ms response times for seamless audio.
- Built modular local server architecture supporting YouTube Music and Spotify integration with privacy-first on-device AI processing.

**LinkinJMS – High-Performance Message Queue Connector** GitHub
*Apache Spark, Scala, JMS, ActiveMQ, AMQP*

- Engineered high-performance ActiveMQ connector achieving sub-50ms message latency with enhanced AMQP security and TLS encryption.
- Implemented bidirectional message streaming for ActiveMQ topics and queues with exactly-once delivery guarantees and automatic failover.